

09/09/2025: Boolean Algebra

CSCI 246: Discrete Structures

Textbook reference: Sec. 7, Scheinerman

Quiz return method

Quizzes are grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Today's Agenda

- Quiz updates (≈ 5 mins)
- Review counterexamples (≈ 15 mins)
- Boolean algebra (≈ 15 mins)
- Group exercises (≈ 15 mins)

Quiz Updates

Review of Counterexamples

Boolean Algebra

Weekly Quiz #1: Scores

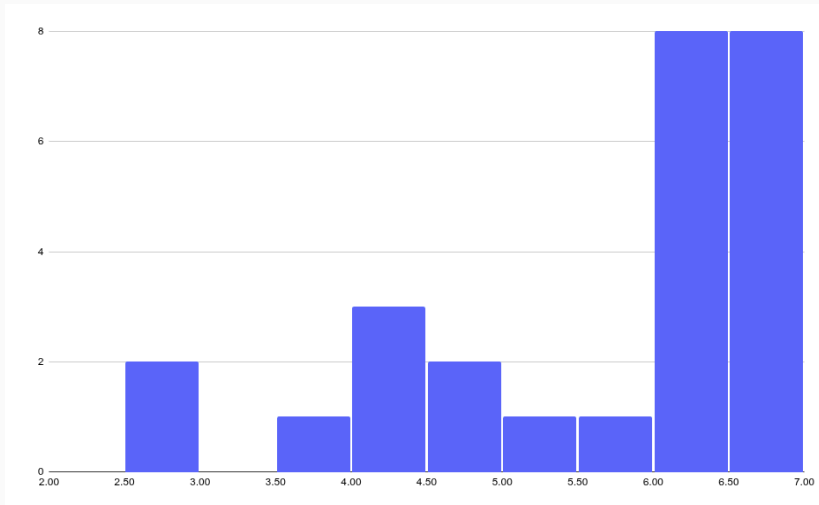


Figure 1: Reading Quiz Scores. Median: 6/7 (85.7%). See repo for feedback.

Study Guide For Quiz 2

Readings (1 question)

Sec 5 (Proofs)

- Know how to prove Props 5.2, 5.3, 5.5, and 5.6.

Sec. 6 (Counterexample)

- Know how to apply Proof Template 3.
- In particular, know how to disprove Statement 6.1.

Group exercises (2 questions)

Know how to do all of the group exercises from these 2 sections.

Quiz Updates

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Boolean Algebra

Review of Sec. 6 (Counterexamples)

Disprove the following conjecture: *Let a and b be integers. If $a|b$ and $b|a$, then $a = b$.*

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Is $a = 1, b = 0$ a valid counterexample?

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Recall. Here's the truth table for unidirectional implication.

		If X, then Y $X \Rightarrow Y$	
X	Y		
T	T	T	
T	F	F	← counterexamples live here
F	T	T	
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Recall. Here's the truth table for unidirectional implication.

		If X , then Y $X \implies Y$	
X	Y	$X \implies Y$	
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T	F	F	← counterexamples live here
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Interpretation. So we need to find a case where proposition $X = \{a|b \text{ and } b|a\}$ is true but proposition $Y = \{a = b\}$ is false.

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Reminder of Definition 3.2 (**Divisible**)

Let a and b be integers. We say that a is *divisible* by b provided there is an integer c such that $bc = a$. The notation for this is $b|a$.

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Solution to poll: The proposition $a|b$ ($= 1|0$) is true, since there is an integer c (namely $c = 0$) such that $ac = b$ (since $1 \cdot 0 = 0$). However, the proposition $b|a$ ($= 0|1$) is false, since there is no integer d such that $bd = a$ (that is, there is no integer d such that $0 \cdot d = 1$). Overall, the hypothesis doesn't hold, so we have not disproven the statement.

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Disprove the following conjecture: *Let a and b be integers. If $a|b$ and $b|a$, then $a = b$.*

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Main Text

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High-level structure

Main Text

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Unravel defn.

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Substituting for a and b , we need to show that there are integers x and y such that $5x = -5$ and $-5y = 5$. We see these equations hold by taking $x = -1$ and $y = -1$.

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Hence $5|-5$ and $-5|5$, so the hypothesis is met.

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This is immediately clear.

Review: how to give a counterexample

Unidirectional implication: $X \implies Y$

Exhibit **one** case in which

- the hypothesis X **holds**, and
- the conclusion Y **fails**.

To check either one, you must **unravel the definitions** inside X and Y .

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What do you have to show to give a counterexample to a bidirectional implication: a proposition of the form $X \iff Y$?

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What do you have to show to give a counterexample to a bidirectional implication: a proposition of the form $X \iff Y$?

Solution

Show that **either** direction fails: **either** that $X \implies Y$ fails, **or** that $Y \implies X$ fails.

Poll

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$$-5|5 = -1$$

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Solution to poll: The statement is **incorrect**! Why?

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Solution to poll: The statement is **incorrect**! Why? $-5|5$ is a **proposition**, so it can't equal -1 . We would instead write just $-5|5$, or perhaps $-5|5 = \text{TRUE}$.

Quiz Updates

Review of Counterexamples

Boolean Algebra

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What is algebra?

Algebra lets us reason about **numbers**. For example, using algebra, we can show that

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What is Boolean algebra?

Boolean Algebra lets us reason about **propositions**. For instance, using Boolean Algebra, we can show that

- If A, then B
- (not A) or B

mean the same thing.

Reasoning about propositions

To reason about propositions, we can use two tools:

- Truth Tables
- Boolean Algebra Properties

Tool #1: Truth Tables

Practice Reading Quiz (Sec. 7 - Boolean Algebra)

Use a truth table to prove that the expressions $x \implies y$ and $(\neg x) \vee y$ are logically equivalent.

Notation reminder

- $\implies$ means implies
- $\neg$ means not
- $\vee$ means or

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Solution. The 3rd and 6th columns are the same.

X	Y	$X \implies Y$	$\neg X$	Y	$(\neg X) \vee Y$
T	T	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	F	T

Tool #2: Properties

In the group exercises, we will see that we can also use the following properties to reason about propositions.

Theorem 7.2

- $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$. (Commutative properties)
- $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ and $(x \vee y) \vee z = x \vee (y \vee z)$. (Associative properties)
- $x \wedge \text{TRUE} = x$ and $x \vee \text{FALSE} = x$. (Identity elements)
- $\neg(\neg x) = x$.
- $x \wedge x = x$ and $x \vee x = x$.
- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. (Distributive properties)
- $x \wedge (\neg x) = \text{FALSE}$ and $x \vee (\neg x) = \text{TRUE}$.
- $\neg(x \wedge y) = (\neg x) \vee (\neg y)$ and $\neg(x \vee y) = (\neg x) \wedge (\neg y)$. (DeMorgan's Laws)

Figure 2: Boolean Algebra Properties

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These properties are proven via truth tables. You can use them as **shortcuts**.



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Boolean Algebra Properties: Example

DeMorgan's Laws

For propositions P and Q :

$$1. \quad \neg(P \wedge Q) = (\neg P) \vee (\neg Q)$$

$$2. \quad \neg(P \vee Q) = (\neg P) \wedge (\neg Q)$$

Poll

How can we understand/interpret these?

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Verbalization

1. "Not (P and Q)" is the same as "(Not P) or (Not Q)."
2. "Not (P or Q)" is the same as "(Not P) and (Not Q)."

Boolean Algebra Properties: Example

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For propositions P and Q :

$$1. \neg(P \wedge Q) = (\neg P) \vee (\neg Q)$$

$$2. \neg(P \vee Q) = (\neg P) \wedge (\neg Q)$$

People use these all of the time in conversation!

Law 1 “You can’t **have your cake** **and** **eat it too**.”

Means: either you don’t **have it**, **or** you don’t **eat it**. That is $\neg(P \wedge Q) = (\neg P) \vee (\neg Q)$.

Law 2 “I don’t **drink** **or** **smoke**.”

Nobody hears that as “or”. Everyone hears “I don’t **drink** **and** I don’t **smoke**”. That is, $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$, applied without noticing.

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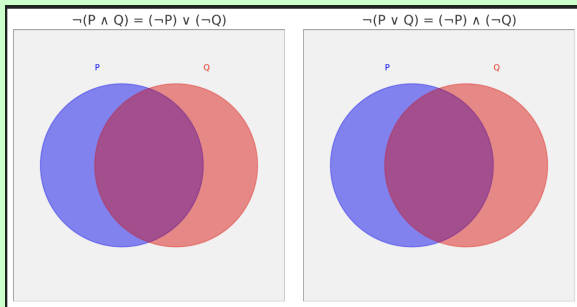
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Visual intuition



- On the left, $\neg(P \wedge Q)$: everything except the overlapping middle part.
- On the right, $\neg(P \vee Q)$: everything outside both circles.